

Data Analysis: Automobiles Dataset

2025/2026

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December 2025

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1 Dataset Description

1.1 Structure

The dataset contains 392 automobile examples produced between 1970 and 1982. This period corresponds to a crucial era for the automotive industry, marked by oil crises that led manufacturers to rethink the fuel efficiency of their vehicles.

The dataset variables are:

- **Performance-related:** miles per gallon (mpg), horsepower, acceleration
- **Engine-related:** cylinders, displacement
- **Structure-related:** weight (in lbs)
- **Contextual:** model year, origin, car name

All numerical variables are of type `float64` or `int64`, except for the car name which is of type `object`. No missing values were detected in the dataset.

1.2 Values

1.2.1 Main Quantitative Variables

Table 1: Descriptive statistics of quantitative variables

Variable	Mean	Median	Std	Min	Max
Miles per gallon	23.45	22.75	7.81	9.00	46.60
Cylinders	5.47	4.00	1.71	3.00	8.00
Displacement	194.41	151.00	104.64	68.00	455.00
Horsepower	104.47	93.50	38.49	46.00	230.00
Weight (lbs)	2977.58	2803.50	849.40	1613.00	5140.00
Acceleration	15.54	15.50	2.76	8.00	24.80
Model year	75.98	76.00	3.68	70.00	82.00

The data shows significant variability, particularly for displacement (std: 104.64) and weight (std: 849.40 lbs). This heterogeneity suggests the presence of very different vehicles in the dataset, ranging from small economy cars to large American sedans.

The average fuel efficiency is 23.45 mpg with a substantial standard deviation (7.81), indicating high dispersion. The minimum value (9 mpg) and maximum (46.6 mpg) show a considerable gap between the least and most efficient vehicles.

1.2.2 Vehicle Origin

Table 2: Distribution of vehicles by origin

Origin	Count	Percentage
USA	245	62.50%
Japan	79	20.15%
Europe	68	17.35%

The dataset is dominated by American vehicles (62.5%), reflecting the US automotive market of that era. Japanese and European vehicles represent 20.15% and 17.35% of the dataset respectively.

Given the observations on quantitative variables and vehicle origin, the dataset presents significant diversity, both in terms of technical characteristics and geographical distribution. This diversity is essential for an in-depth analysis of factors influencing fuel efficiency.

2 Dataset Analysis

2.1 Correlations

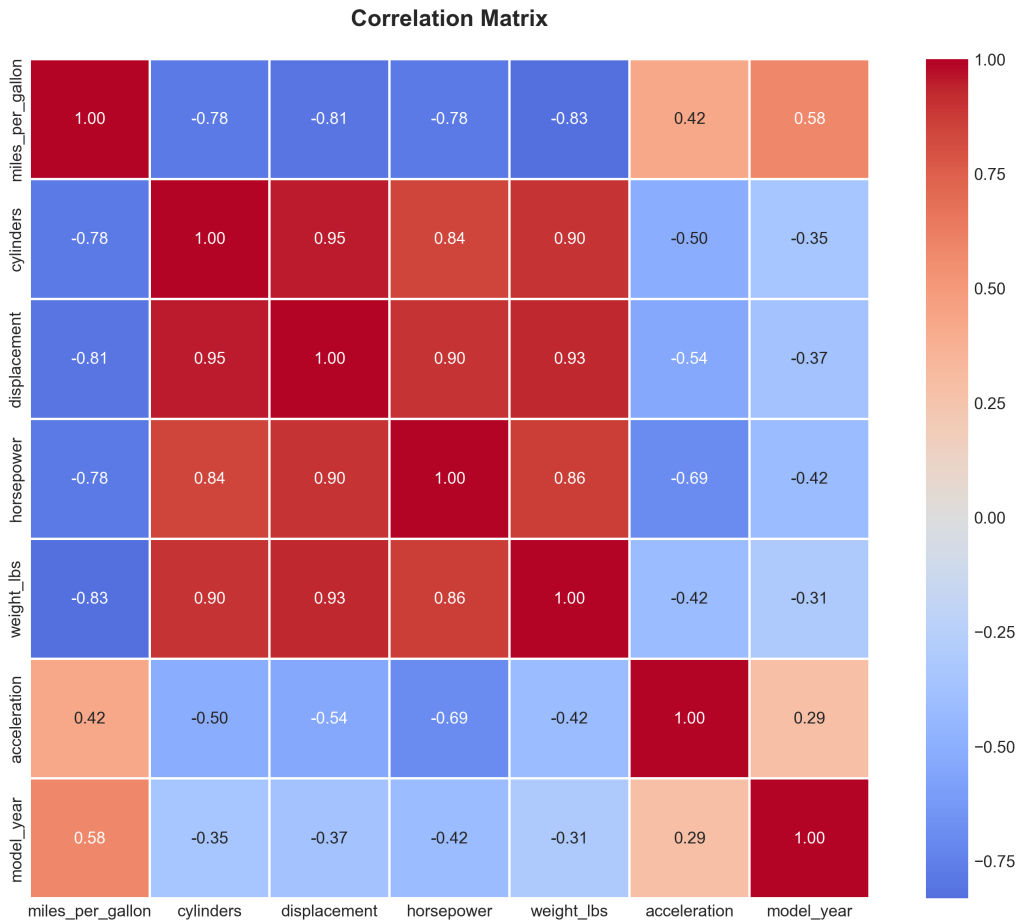


Figure 1: Correlation matrix of quantitative variables

Based on the correlation matrix (Figure 1):

- The strongest correlation with MPG is **negative**, with weight (-0.83). The heavier a car, the less fuel efficient it is.
- Displacement also shows a strong negative correlation with MPG (-0.81), suggesting that large-displacement engines consume more fuel.
- Horsepower and number of cylinders are both strongly negatively correlated with MPG (respectively -0.78 and -0.78).
- Acceleration shows a moderate positive correlation (+0.42) with MPG, indicating that more efficient cars tend to have better acceleration (lighter cars).
- Model year shows a positive correlation (+0.58) with MPG, confirming an improvement in efficiency over the years.

Variables related to engine physical characteristics (cylinders, displacement, horsepower, weight) are highly intercorrelated (correlations above 0.89), suggesting strong multicollinearity. This indicates that a powerful engine generally implies a heavy vehicle with large displacement.

2.2 Data Distribution

Table 3: Average characteristics by origin

Variable	USA	Europe	Japan
Miles per gallon	20.03	27.60	30.45
Horsepower	119.05	80.56	79.84
Weight (lbs)	3372.49	2433.47	2221.23
Acceleration	14.99	16.79	16.17
Displacement	247.51	109.63	102.71
Cylinders	6.28	4.16	4.10

Analysis by origin reveals marked differences:

- Japanese cars are the most efficient (30.45 mpg on average), followed by European cars (27.60 mpg). American cars are significantly less efficient (20.03 mpg).
- American vehicles are heavier (3372 lbs on average) compared to Japanese (2221 lbs) and European (2433 lbs) cars.
- American cars have on average more power (119 hp) and larger displacement (247.51) than their European and Japanese counterparts.
- This difference is explained by American market preferences for larger and more powerful vehicles, before CAFE standards were imposed.

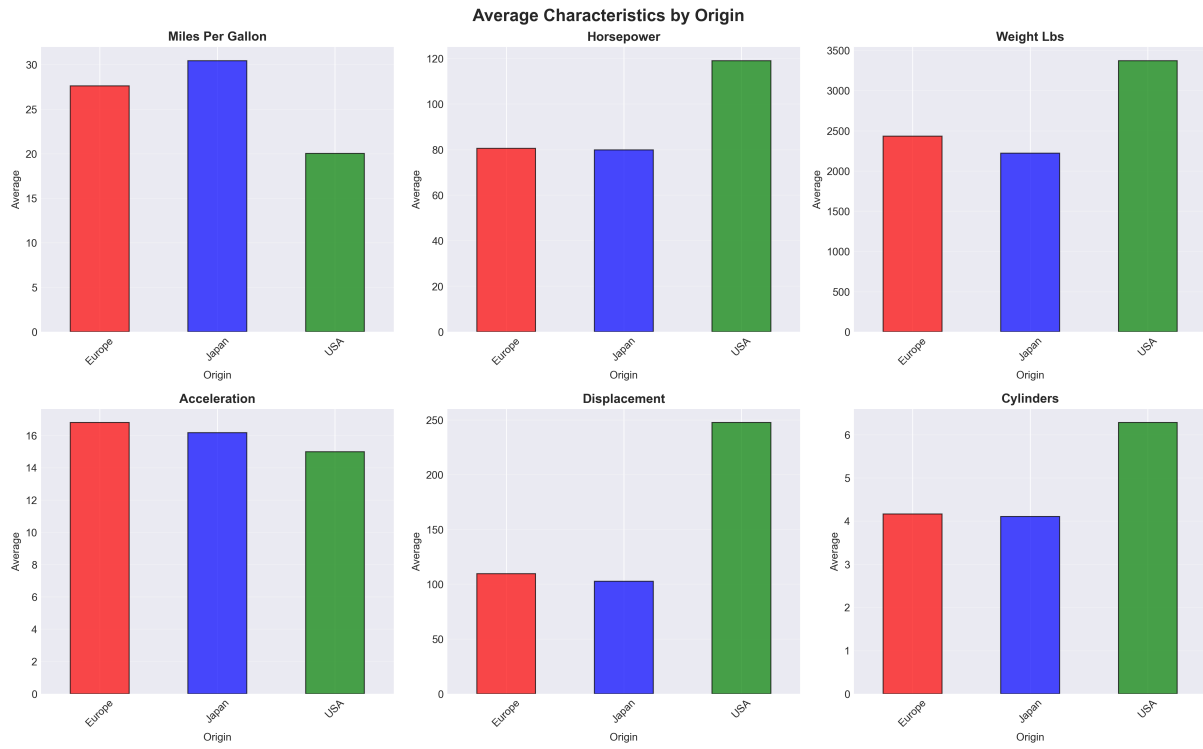


Figure 2: Comparison of key characteristics by origin

2.3 Distributions

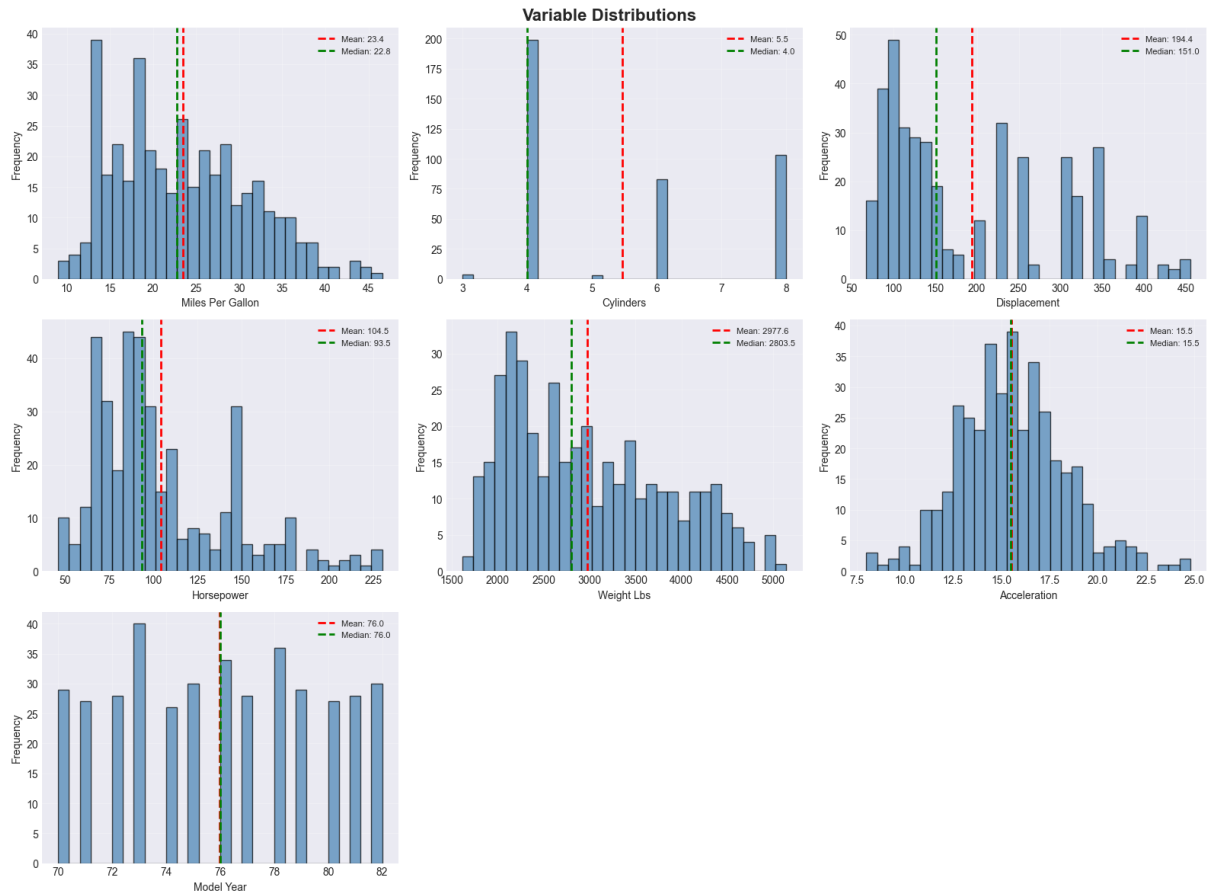


Figure 3: Distribution of quantitative variables

Variable distributions (Figure 3) show:

- **Miles per gallon:** slightly right-skewed distribution, with concentration around 20-25 mpg.
- **Cylinders:** bimodal distribution with peaks at 4, 6 and 8 cylinders, reflecting three types of engines.
- **Displacement and Horsepower:** asymmetric distributions with long tails toward high values, indicating the presence of some very powerful vehicles.
- **Weight:** relatively symmetric distribution around the median (2803 lbs).
- **Model year:** uniform distribution over the 1970-1982 period, showing balanced representation of each year.

Means and medians are generally close for most variables, suggesting relatively symmetric distributions, except for displacement and horsepower which show extreme values.

2.4 CAFE Standard Compliance

In 1982, the Corporate Average Fuel Economy (CAFE) standard required vehicles to achieve an average of 24 mpg. Compliance analysis reveals:

- **181 out of 392 cars (46.2%)** meet or exceed the 24 mpg threshold.
- **211 cars (53.8%)** do not meet this standard.

Table 4: Average characteristics according to CAFE compliance

Variable	< 24 mpg	24 mpg	Difference (%)
Miles per gallon	17.37	30.53	+75.7%
Cylinders	6.62	4.13	-37.6%
Displacement	264.79	112.37	-57.6%
Horsepower	127.44	77.69	-39.0%
Weight (lbs)	3558.46	2300.43	-35.4%
Acceleration	14.73	16.49	+11.9%
Model year	74.52	77.69	+4.3%

Cars compliant with the CAFE standard have distinct characteristics:

- **35.4% lighter** on average (2300 lbs vs 3558 lbs)
- **57.6% less displacement** (112 vs 265 cubic inches)
- **39% less power** (78 hp vs 127 hp)
- Slightly more recent (model year 77.69 vs 74.52)

Analysis by origin shows that:

- **83% of Japanese cars** meet the CAFE standard
- **74% of European cars** meet the standard
- Only **27% of American cars** meet the standard

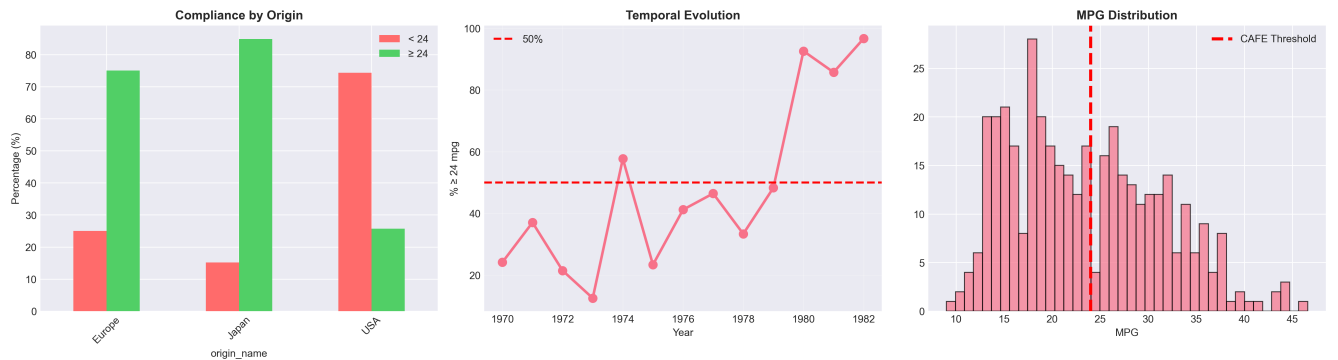


Figure 4: CAFE compliance analysis: by origin, temporal evolution, and distribution

The temporal evolution (Figure 4) shows a clear improvement: the percentage of cars meeting the standard increases from about 20% in 1970 to over 80% in 1982, demonstrating the effectiveness of regulations and technological innovation in the automotive industry in response to oil crises.

2.5 Principal Components Analysis

Table 5: Variance explained by principal components

Component	Explained Variance	Cumulative Variance
PC1	71.58%	71.58%
PC2	12.37%	83.95%
PC3	10.41%	94.35%
PC4	2.63%	96.98%
PC5	1.74%	98.72%
PC6	0.78%	99.50%
PC7	0.50%	100.00%

Principal component analysis reveals that:

- **PC1 captures 71.58% of variance**, indicating that one main dimension dominates the dataset.
- **The first 2 components explain 83.95%** of total variance, allowing efficient 2D representation.
- **3 components are sufficient to capture over 90%** of variance, suggesting the dataset can be effectively reduced.

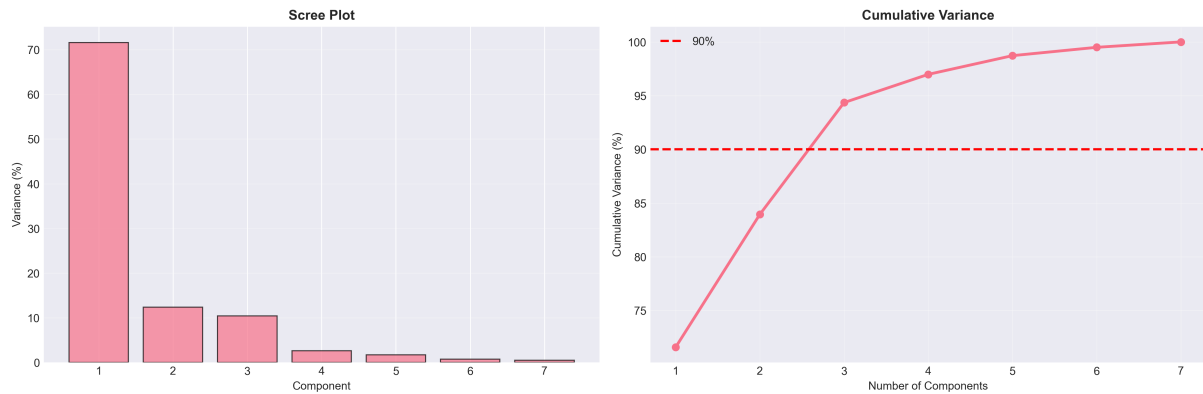


Figure 5: Explained variance and cumulative variance by principal component

2.5.1 Interpretation of Principal Components

Table 6: Variable contributions to the first two principal components

Variable	PC1	PC2
Miles per gallon	-0.398	0.207
Cylinders	0.416	0.199
Displacement	0.429	0.180
Horsepower	0.423	0.085
Weight	0.414	0.225
Acceleration	-0.285	-0.007
Model year	-0.230	0.910

PC1 (71.58% variance) mainly represents vehicle physical characteristics and performance:

- Strong positive contributions: displacement (0.429), horsepower (0.423), cylinders (0.416), weight (0.414)
- Negative contribution: miles per gallon (-0.398)
- **Interpretation:** PC1 opposes heavy, powerful, inefficient vehicles to light, efficient ones.

PC2 (12.37% variance) is dominated by the temporal dimension:

- Major contribution: model year (0.910)
- **Interpretation:** PC2 captures technological evolution over time.

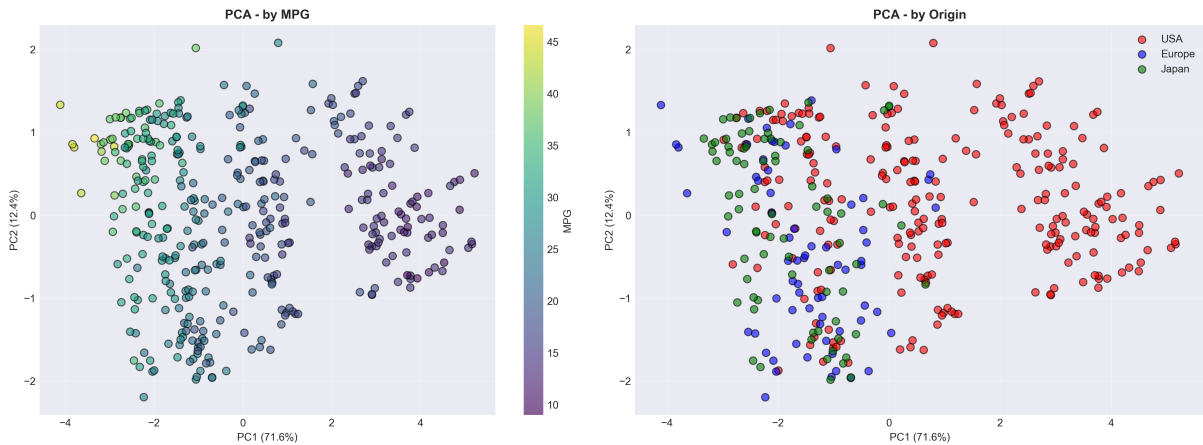


Figure 6: Data projection on the first two principal components

The 2D projection (Figure 6) shows:

- Clear separation between efficient vehicles (high MPG, left) and inefficient ones (right)

- American cars concentrate mostly on the right (inefficient)
- Japanese and European cars dominate the left side (efficient)
- PC2 captures temporal evolution, with more recent cars toward the top

The 2D representation via PCA effectively visualizes the dataset structure while preserving 84% of information. This dimensionality reduction confirms that vehicle characteristics are highly related and can be summarized by a few main dimensions.

2.6 Regression Analysis for Fuel Efficiency Prediction

To predict fuel efficiency (miles per gallon), we tested several regression approaches with different levels of complexity. The data was split into training (80%, 313 examples) and test (20%, 79 examples) sets.

2.6.1 Model Comparison

Table 7: Performance of regression models

Model Type	Description	R ² Test	RMSE Test (mpg)
Simple Linear	Weight only	0.6533	4.21
Polynomial Degree 2	Weight (quadratic)	0.6730	4.09
Polynomial Degree 3	Weight (cubic)	0.6730	4.09
Multiple Linear	5 variables	0.6485	4.24

2.6.2 Key Findings

- **Polynomial regression (degree 2) achieves the best performance** with an R² of 0.6730 and RMSE of 4.09 mpg.
- The quadratic model captures a non-linear relationship between weight and fuel efficiency that the simple linear model misses.
- Interestingly, the cubic model (degree 3) shows **no improvement** over the quadratic model (same R² and RMSE), suggesting that second-order terms are sufficient.
- The **multiple linear regression model with 5 variables** performs slightly worse than the simple weight-only model, likely due to **multicollinearity** among predictors (weight, displacement, horsepower, and cylinders are highly correlated).

2.6.3 Optimal Model: Quadratic Regression (Degree 2)

The quadratic regression model with weight as predictor provides the best balance of complexity and performance. The model captures a non-linear relationship where MPG decreases with weight, but the rate of decrease slows down for heavier vehicles.

Key performance metrics:

- The quadratic model explains **67.30% of MPG variance**, a 2% improvement over the linear model

- The average prediction error decreases from 4.21 to 4.09 mpg
- The model shows that weight is the most important predictor, capturing the essential relationship while avoiding multicollinearity issues

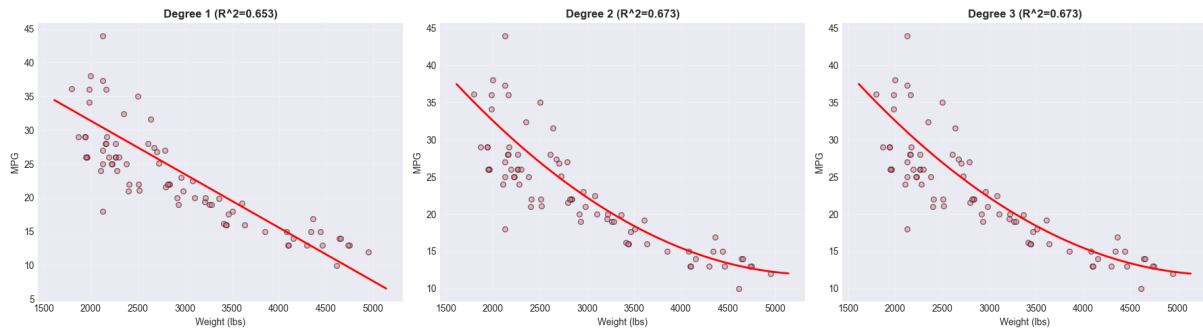


Figure 7: Comparison of polynomial regression models (degrees 1, 2, and 3)

Figure 7 clearly shows:

- The linear model (degree 1) represents a straight-line relationship
- The quadratic model (degree 2) captures a curved relationship that better fits the data distribution
- The cubic model (degree 3) provides no visible improvement over the quadratic model

2.6.4 Model Selection Recommendation

Based on the principle of parsimony and performance:

1. **Recommended model: Quadratic regression (degree 2)** - best performance with reasonable complexity
2. **Alternative: Simple linear regression** - nearly as good for many practical purposes
3. **Not recommended: Multiple linear regression** - suffers from multicollinearity without performance benefit

The quadratic relationship confirms that weight is the primary determinant of fuel efficiency, with a non-linear effect where very heavy vehicles show a less rapid decline in efficiency compared to medium-weight vehicles.

3 Conclusion

In conclusion, this analysis of the automobile dataset from the late 1970s and early 1980s reveals significant insights into the factors influencing vehicle fuel efficiency during a period of major technological transition. Weight emerges as the strongest single predictor of miles per gallon, showing a consistent and dominant negative relationship with fuel efficiency across all vehicle types.

Clear geographical patterns emerge, with Japanese and European manufacturers producing significantly more fuel-efficient vehicles than their American counterparts. These differences reflect distinct design philosophies and engineering approaches.

The Corporate Average Fuel Economy (CAFE) standard of 24 mpg provides a useful benchmark for assessing progress. While only a minority of vehicles in the dataset meet this threshold, the data shows remarkable improvement over the 1970-1982 period, indicating successful adaptation to new regulatory requirements.

Technical characteristics such as displacement, horsepower, and number of cylinders show strong multicollinearity, suggesting that these features capture overlapping aspects of vehicle design. This redundancy explains why simple models based only on weight perform nearly as well as more complex multivariate approaches.

Principal Component Analysis confirms that vehicle characteristics are highly interrelated, with the first component alone explaining over 71% of variance by opposing heavy, powerful vehicles against lighter, more efficient ones. The second component captures temporal evolution, showing clear technological progress throughout the studied period.

Overall, the analysis demonstrates that while linear relationships provide a good starting point for understanding fuel efficiency, the quadratic regression model reveals a more nuanced reality. Vehicle weight is not simply a linear constraint but has a complex effect that moderates at higher weights, suggesting that engineering solutions become more sophisticated as vehicles grow heavier.

4 Annexes

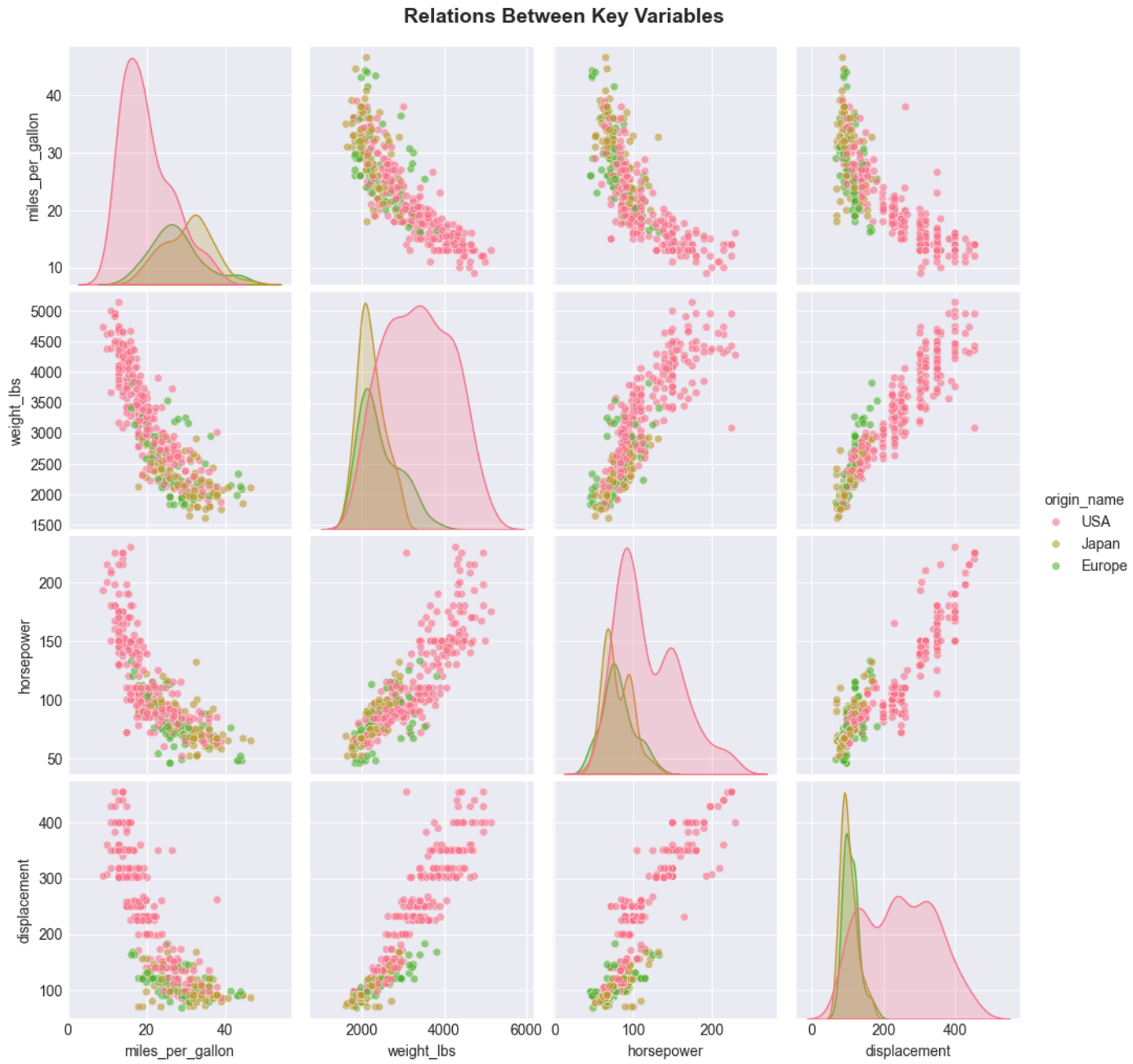


Figure 8: Pairplot of the data with origin name